Characterization Report: Sky130 Standard Cell Library

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Course: VLSI Design (Project 1)

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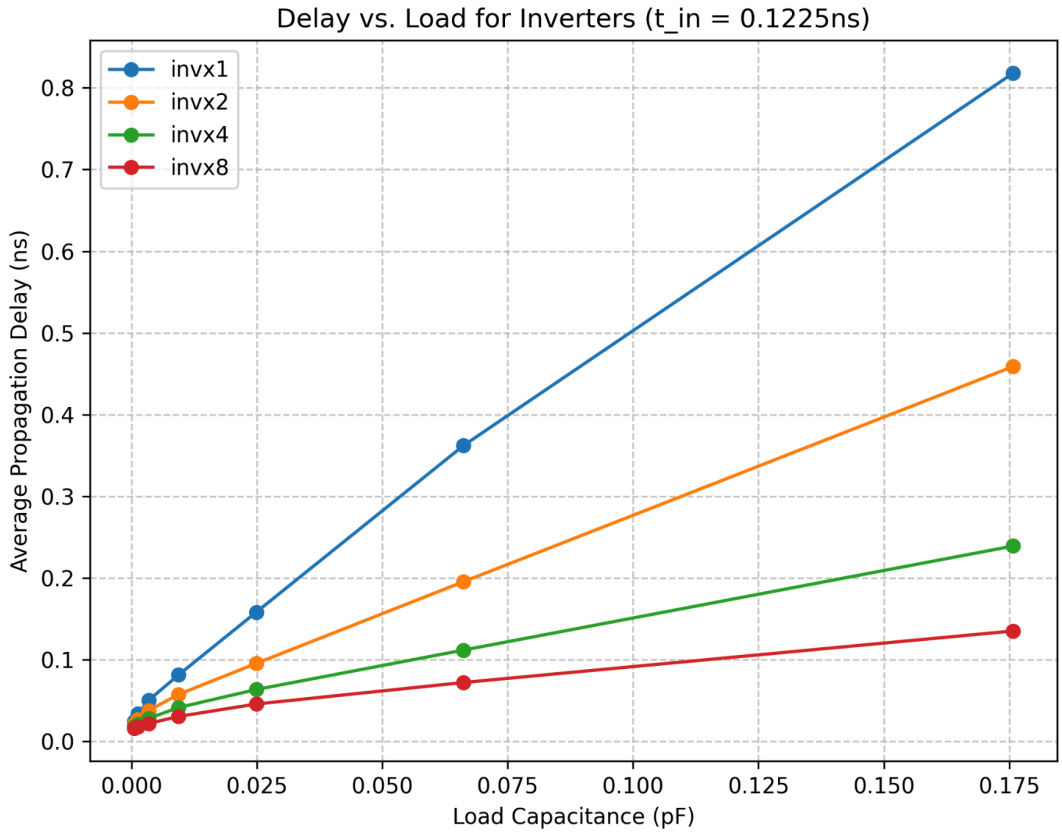
1. Introduction

This report details the characterization of a custom standard cell library using the SkyWater 130nm process (SKY130). We designed 13 fundamental digital cells, sized them for balanced rise/fall transitions, and developed a Python-based automation suite to generate Non-Linear Delay Model (NLDM) tables.

2. Delay vs. Load Analysis

The following plot illustrates the average propagation delay as a function of load capacitance for the inverter family. All simulations were performed at a fixed input transition (t_{in}) of **0.1225ns**.



Observations:

- As expected, delay increases linearly with load capacitance (C_{load}).
- Larger cells (e.g., invx8) show a significantly flatter slope compared to invx1, demonstrating their superior "drive strength" when handling high-fanout nets.

3. Analytical Comparison: RC Model vs. SPICE (NLDM)

A critical requirement of this study is comparing the first-order RC delay model against the high-fidelity SPICE results. We focused on the invx1 cell at the library's midpoint: $t_{in} = 0.1225\text{ns}$ and $C_{load} = 0.0094\text{pF}$.

3.1 Theoretical RC Delay Calculation

Using the effective resistance (R_{nmos}) derived from Tutorial 2 measurements (124ps discharge for 25fF):

$$R_{nmos} = \frac{124 \times 10^{-12}}{0.69 \times 25 \times 10^{-15}} \approx 7188 \Omega$$

For the target 0.0094pF load, the theoretical delay is:

$$t_{delay} = 0.69 \times R_{nmos} \times C_{load}$$
$$t_{delay} = 0.69 \times 7188 \times 9.4 \times 10^{-15} \approx 46.6 \text{ ps}$$
Theoretical RC Delay: 0.0466 ns

3.2 SPICE (NLDM) Results

From our automated characterization of the invx1 cell:

Extracted SPICE Delay: 0.08273 ns

3.3 Discussion of Discrepancies

The SPICE-extracted delay (**0.08273 ns**) is approximately **77% higher** than the theoretical RC prediction (**0.0466 ns**). This discrepancy highlights the fundamental limitations of the first-order RC model:

1. **Input Transition Sensitivity:** The RC model assumes a perfect step input. In reality, the **0.1225ns** input slew means the transistors are in a sub-optimal, partially-on state for a significant duration, slowing the output transition.
2. **Short-Circuit Current:** During the input transition, both pull-up and pull-down networks are briefly active simultaneously, creating a "crowbar" current that fights the output change—an effect the RC model ignores.
3. **Conclusion:** While the RC model is useful for quick "back-of-the-envelope" estimates, NLDM tables are essential for modern timing closure because they account for the non-linear dependencies of input slew.

4. Conclusion

The automation script successfully characterized 13 cells across 49 corners each. The resulting NLDM tables provide the necessary timing data for logic synthesis and static timing analysis (STA) in the Sky130 process.